

Machine Learning : Project 1

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Abstract—In this project we had a dataset on which the goal was to find the best way to put together a good predictive model, in this paper we are going to have an overview of the methods we have tried, some of our findings and finally what are the parameters of our model.

I. INTRODUCTION

The dataset is coming from the Behavioral Risk Factor Surveillance System, it is giving us many information on some health related features, our goal is to predict if a given person with some set health-related information will lead to a MIHCD (Coronary heart disease). To classify these information we are going to use various techniques of binary classification being the gradient descent, stochastic gradient descent, least square, ridge regression, logistic regression and regularized logistic regression.

II. PREPROCESSING

We are never guaranteed that the dataset we are given for a specific application is uniform and full, the main goal of the preprocessing is to get to know the dataset, find some problems and their solution mainly we looked into the feature fullness, their variance, their correlation and for categorical features with less than 5 distinct values a way to encode them in a more effective way.

A. Feature fullness

Having some non assigned values to some features is not a real problem as we can infer those gaps using mean or median for example, it becomes more of a problem when our feature is mainly empty and then for most of the person we will just analyze a data we don't already know, here we can see the proportion of features and their ratio of fullness As we can

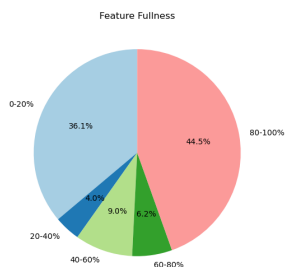


Fig. 1. Feature fullness

see there are a good proportion of the features that are close to be fully populated but nearly 40% of those which are mainly empty for those even if we filled the unassigned data the model

would end up being trained on mostly inferred data that don't carry any meaningful information. For our model we tried multiple threshold of missing value percentage and came up with a value for the model of 40%, if a feature had more than 40% of missing value we got rid of it.

B. Feature variance

In a dataset it is possible that some features have a variance of 0, it happens when every points of a dataset have the same value, when it happens the entropy of the feature is 0 and we were able to remove it without consequences while improving the speed of the model's training.

C. Feature correlation

In a given dataset it is possible that some feature share some trends in them, if a given feature decrease when another increase following a pattern if we keep both it could introduce some bias to the model as it would give more weight to some feature to some other. In our model we decided to remove one feature for every pair of highly correlated features, a feature was considered as highly correlated if the absolute value of the Pearson's correlation we above 0.95.

D. Categorical Features

It is possible that in a dataset there are some categorical feature for which their value doesn't have any meaningful sense of ordering, for them seeing how close they are wouldn't bring anything and for that we had to include one hot encoding a way of encoding categorical features in different features describing if the data point is in this category or not. First to determine if a feature was categorical or not we had to put a threshold on the number of unique value a feature had.

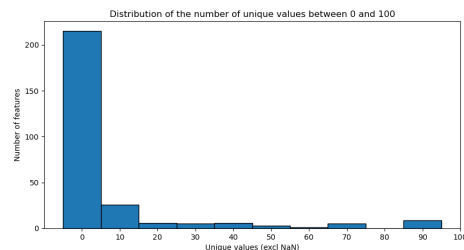


Fig. 2. Caption

As we can see most of the categorical features are below 20 unique values, we will one hot encode only those to avoid an uncontrolled growth of the dimensions. Doing that we keep most of the information gain while still keeping a reasonable training time for the model.

III. MODEL TRAINING

While having a decent data preprocessing pipeline is relevant, it can't express if we train a model with bad hyperparameters, or with a model who can't understand the depth of a dataset. We tried every models on a reduction of the problem with low computation time to first see what were the performance baseline for every models.

Model	Best F1 Score	Execution Time (s)
Linear Regression (GD)	0.1636	5.1
Linear Regression (SGD)	0.2059	0.0
Least Squares Regression	0.2899	0.0
Ridge Regression	0.2899	0.6
Logistic Regression	0.2523	2.9
Regularized Logistic Regression	0.2883	36.0

TABLE I

MODEL COMPARISON BASED ON F1 SCORE AND EXECUTION TIME.

As we can see the performance were We decided to use regularized logistic regression as it has a great possibility of fine tuning. As we used cross validation to find the best value for gamma and lambda the only downside was that it was exponentially longer to cross validate due to needing to optimize both lambda and gamma instead of a single hyperparameter. We chose to some fixed values that covered a wide range while still keeping it as few as possible to avoid falling in the trap of having to wait an hour every time we changed something so we could furthermore optimize our data preprocessing pipeline and decision threshold. The options for gamma and lambda were [0.0001, 0.001, 0.01, 0.1, 0.5, 1] As mentioned above for the hyperparameters we decided to use cross validation with 5 folds. It is a way to ensure that there is less of a bias toward the split of the data while still having a decent training time on the model. After finding the ideal value for gamma we decided to try modifying the value of the decision threshold and seeing what we would get. Moving the threshold from 0.5 to 0.8 got the f1 score from approximately 0.3 to 0.4.

IV. CONCLUSION

During this project we explored the preprocessing methods we can use and several models and how they react to real world data after this work we concluded that regularized logistic regression was able to bring good performance while still being relatively quick but required a small set of possible values for gamma and lambda.

Although the final performance of the model remain modest it is a great example that real world data are way harder than staying on the theoretical side as we had to find a way to handle missing value, choose which feature we kept and which we have to remove for the training. In future work we are looking for seeing more advanced methods of preprocessing and model training being able to better adapt to real world data set and bring better final performances.